

AI Agents & Autonomous Swarms

Multi-Agent Topologies, Reflection Loops, Memory & Enterprise Swarms

AGENT ARCHITECTURE & FRAMEWORKS

1. Autonomous Startup Team Swarm

Design an autonomous team (PM, Engineer, QA, DevOps) collaborating via message brokers with human-in-the-loop gates.

2. Self-Healing Code Execution Loop

Build an automated loop executing code in Docker sandboxes, parsing stack traces, and iteratively fixing bugs.

3. Hierarchical Task Delegation Engine

Implement an orchestrator decomposing complex user goals into sub-agent graphs with state-machine checkpoints.

4. Long-Term Memory & Vector Hybrid Store

Combine episodic vector memory, semantic knowledge graphs, and sliding working memory for long tasks.

SECURITY, GUARDRAILS & FINOPS

5. Semantic Firewall & Injection Shield

Sanitize incoming payloads, detect indirect prompt injections, and enforce strict tool execution boundaries.

6. Agentic Infinite Loop Breakers

Detect circular reasoning and tool failures in real time, forcing strategic pivots or fallback procedures.

7. Multi-Agent Consensus & Debate Protocol

Implement 3 divergent persona agents debating high-stakes decisions with an arbiter judge agent.